

ECSE-4040

Digital Electronics

Lecture 13

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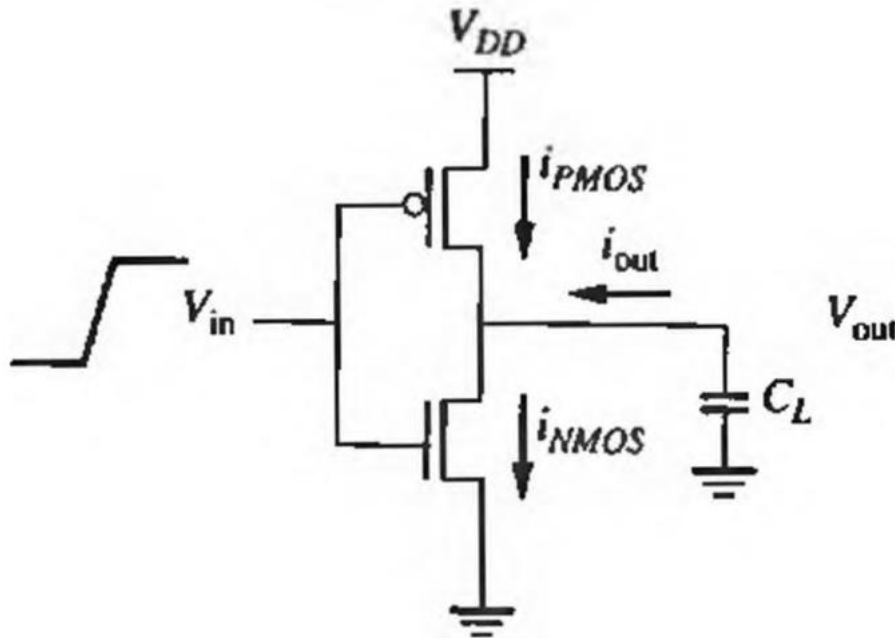
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<https://lms.rpi.edu>

Lecture Outline

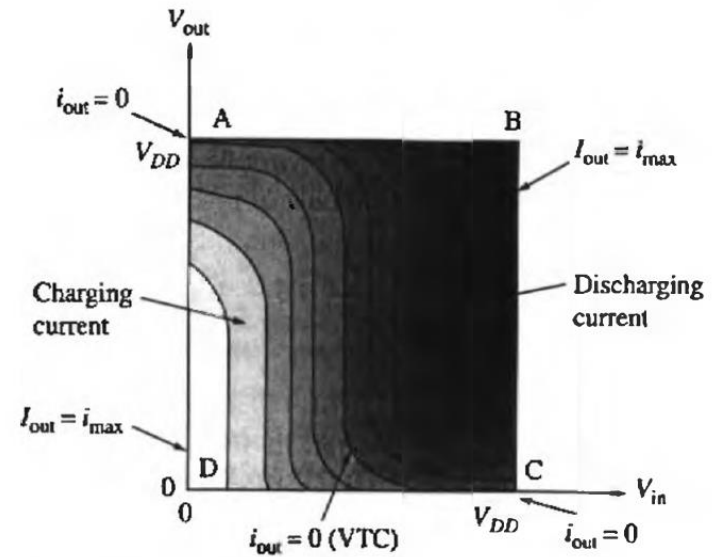
- Input Slope Effects
- Elmore Delay

Input Slope Effects

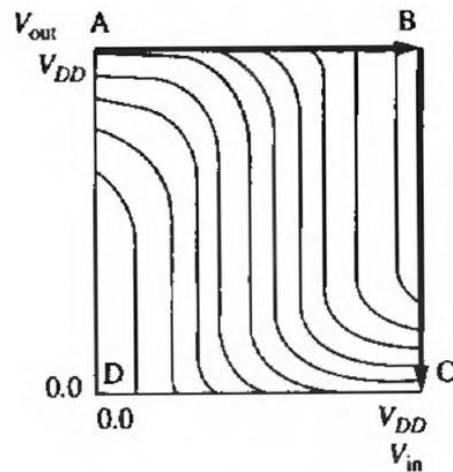
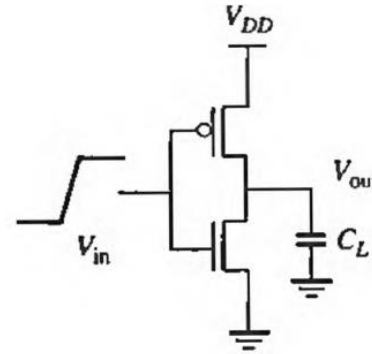
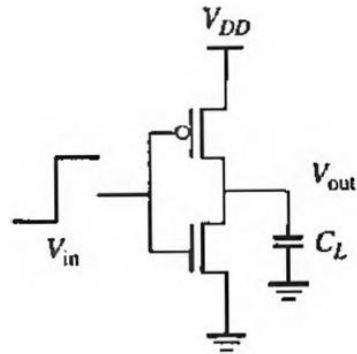


$$i_{out} = C_L \frac{dV_{out}}{dt} = i_{NMOS} - i_{PMOS}$$

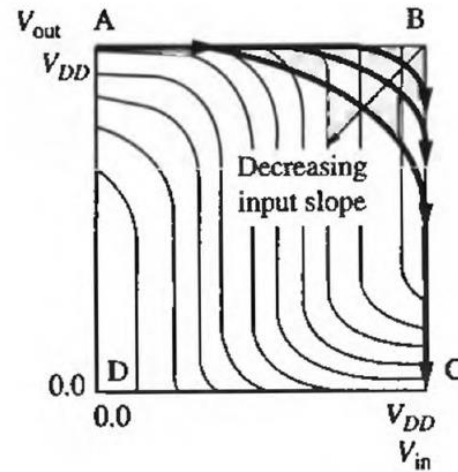
- Not all current from PMOS goes to C_L
- Not all current through NMOS is from C_L
- When V_{in} is transitioning, short circuit current bypasses C_L .



Input Slope Effects

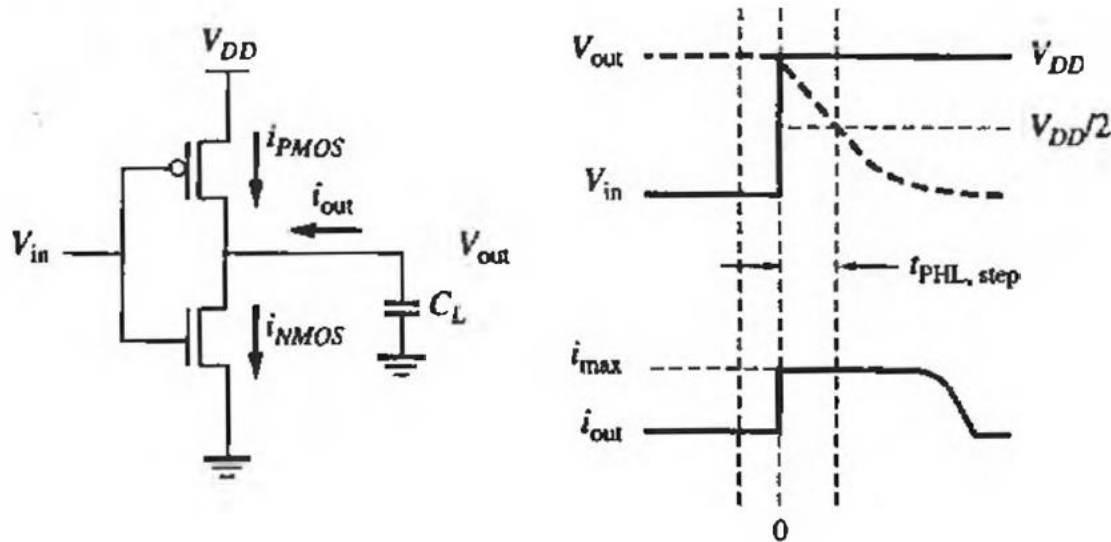


(a) Step input trajectory



(b) Ramp input trajectory

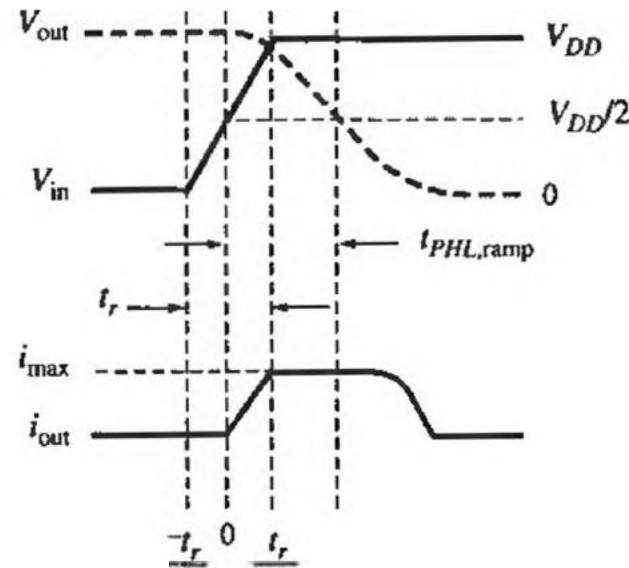
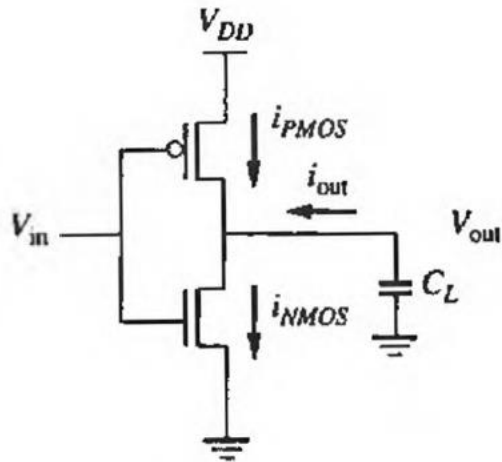
Input Slope Effects



$$i_{out} = i_{max} = C_L \frac{dV_{out}}{dt} = C_L \frac{V_{DD}/2}{t_{PHL, step}}$$

$$\therefore t_{PHL, step} = C_L \frac{V_{DD}/2}{i_{max}}$$

Input Slope Effects



$$i_{out} = C_L \frac{dV_{out}}{dt}$$

$$\therefore \int i_{out} dt = \int C_L dV_{out}$$

$$\therefore \int_0^{t_r/2} i_{max} \frac{t}{t_r/2} dt + \int_{t_r/2}^{t_{PHL,ramp}} i_{max} dt = C_L \int_0^{V_{DD}/2} dV_{out}$$

$$\therefore \frac{i_{max}}{t_r/2} \frac{t^2}{2} \Big|_0^{t_r/2} + i_{max} t \Big|_{t_r/2}^{t_{PHL,ramp}} = C_L V_{DD}/2$$

$$\therefore i_{max} \frac{t_r}{4} + i_{max} (t_{PHL,ramp} - t_r/2) = C_L V_{DD}/2$$

$$\therefore t_{PHL,ramp} = \frac{t_r}{4} + \frac{C_L V_{DD}/2}{i_{max}}$$

Input Slope Effects

- Integration of KCL yields the following:

$$t_{PHL,ramp} = \frac{t_r}{4} + t_{PHL,step}$$

- Assuming $1/4^{\text{th}}$ rise time ($10\% \rightarrow 90\%$) = $1/2$ propagation delay ($100\% \rightarrow 50\%$) ($t_r \cong 2t_{PLH,step}$)

$$\begin{aligned} t_{PHL,ramp} &\approx \frac{2t_{PLH,step}}{4} + t_{PHL,step} \\ &= \frac{t_{PHL,step}}{2} + t_{PHL,step} \end{aligned}$$

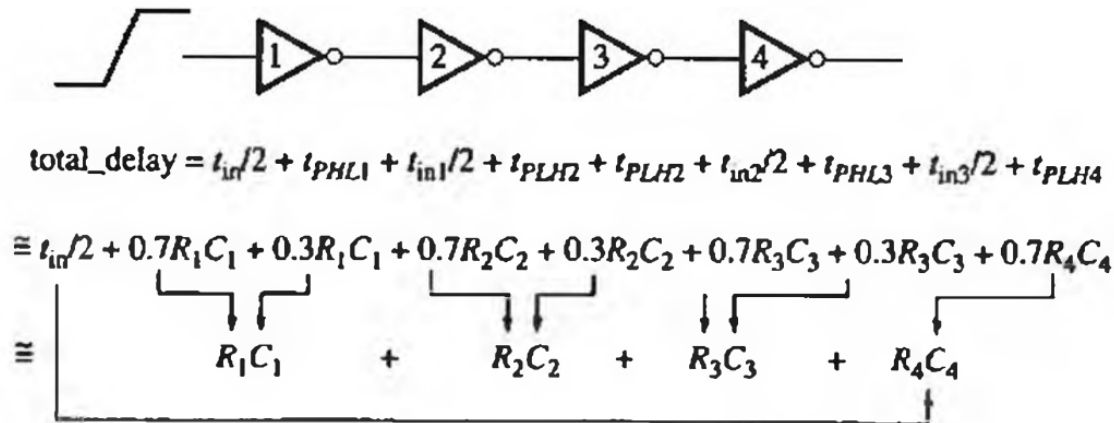
Input Slope Effects

- For an initial fixed input slope:

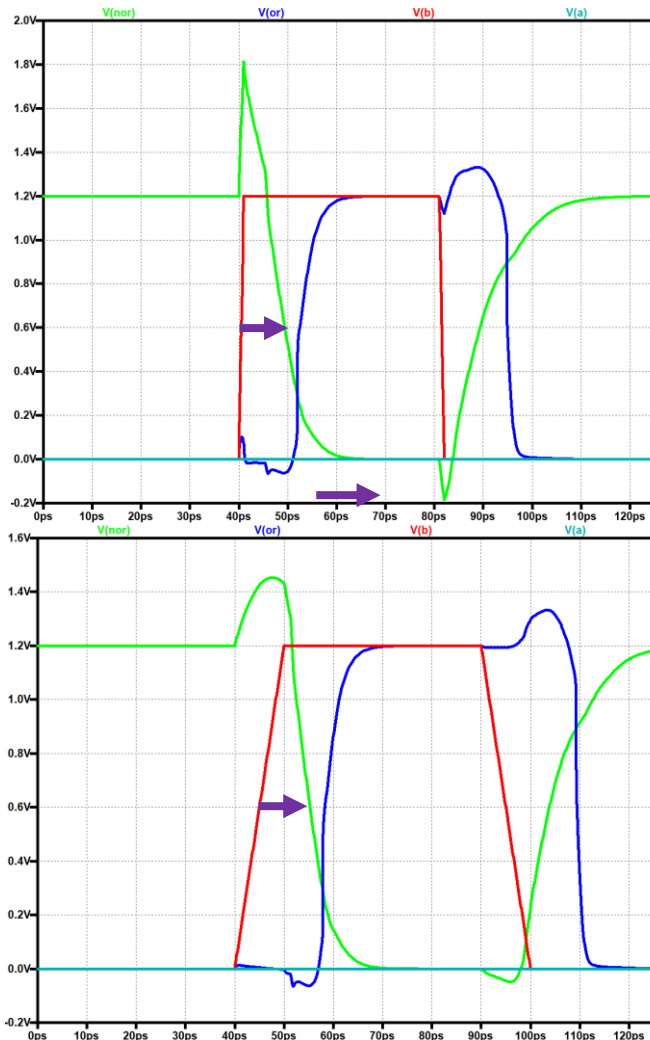
$$t_{\text{ramp}} \approx \frac{t_{\text{in}}}{2} + t_{\text{step}}$$

- For an inverter chain (rapid series of output transition):

$$\text{Total_delay} \approx \sum_i R_i C_i$$



Input Slope Effects



- $\tau_{\text{PHL}}(\tau_{\text{in}}=1\text{ps}) = 8.87\text{ ps}$

$$\tau_{\text{in}} = 1\text{ps}, \frac{\tau_{\text{in}}}{2} = 0.5\text{ps}$$

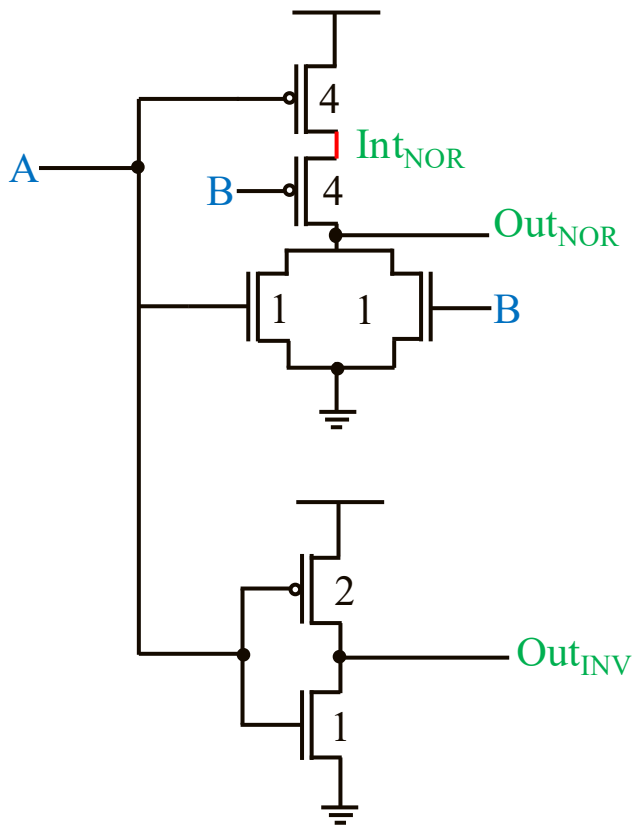
- $\tau_{\text{PHL}}(\tau_{\text{in}}=10\text{ps}) = 10.23\text{ ps}$

$$\tau_{\text{in}} = 10\text{ps}, \frac{\tau_{\text{in}}}{2} = 5\text{ps}$$

- 4.5 ps $\tau_{\text{in}}/2$ difference
- Only 1.36 ps τ_{PHL} change
- Feedthrough current

Internode Delay

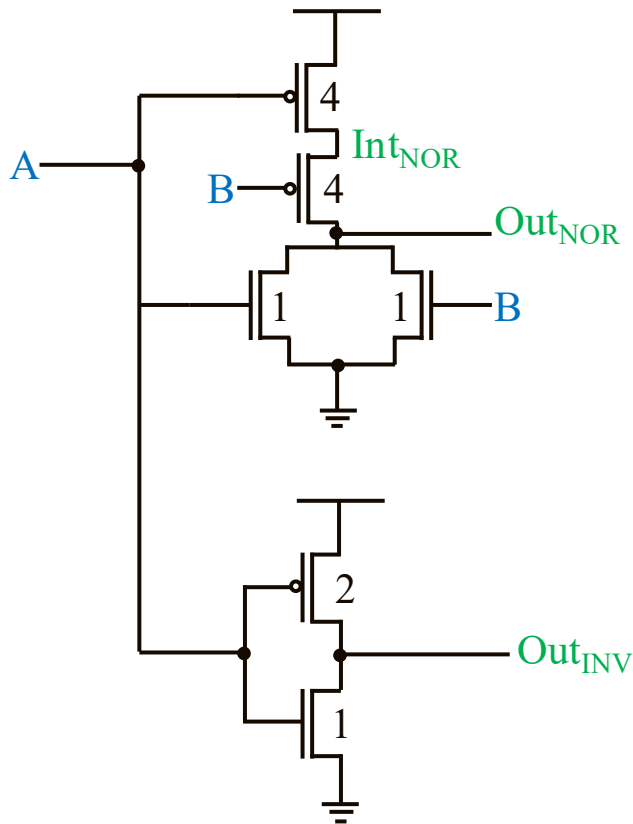
Recall that in Lecture 13 we ignored the internode PMOS delay:



$$C_{NOR(int)} = 4 \cdot 0.1f = 0.4fF \leftarrow \text{Node between PMOS}$$

- Out_{NOR} cannot charge until Int_{NOR} is charged first.
- Charging Int_{NOR} only requires passing current through the upper PMOS, not both PMOS in series.
 - Charges with half the PMOS branch resistance

Internode Delay



$$C_{output} = C_{self} = (W_n + W_p) \cdot 0.1 \text{ fF/100nm}$$

$$C_{NOR} = [(1+1)+4] \cdot 0.1 \text{ fF} = 0.6 \text{ fF}$$

$$C_{NOR(int)} = 4 \cdot 0.1 \text{ fF} = 0.4 \text{ fF} \leftarrow \text{Node between PMOS}$$

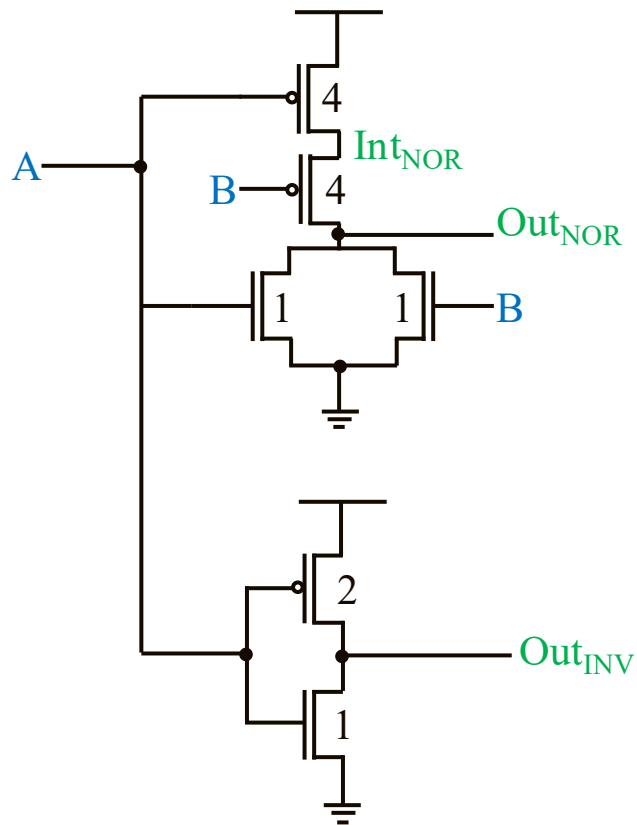
$$R(Out_{NOR}) = 30 \text{ k}\Omega \cdot \frac{L}{W} = 30 \text{ k}\Omega \cdot \frac{2}{4} = 15 \text{ k}\Omega$$

$$R(Int_{NOR}) = 30 \text{ k}\Omega \cdot \frac{L}{W} = 30 \text{ k}\Omega \cdot \frac{1}{4} = 7.5 \text{ k}\Omega$$

$$\begin{aligned} \tau_{pLH} &= 0.7 \cdot RC = 0.7 \cdot (R_{INT} \cdot C_{INT} + R_{OUT} \cdot C_{OUT}) \\ &= 0.7 \cdot (7.5 \text{ k}\Omega \cdot 0.4 \text{ fF} + 15 \text{ k}\Omega \cdot 0.6 \text{ fF}) \\ &= 0.7(3 \text{ p} + 9 \text{ p}) \end{aligned}$$

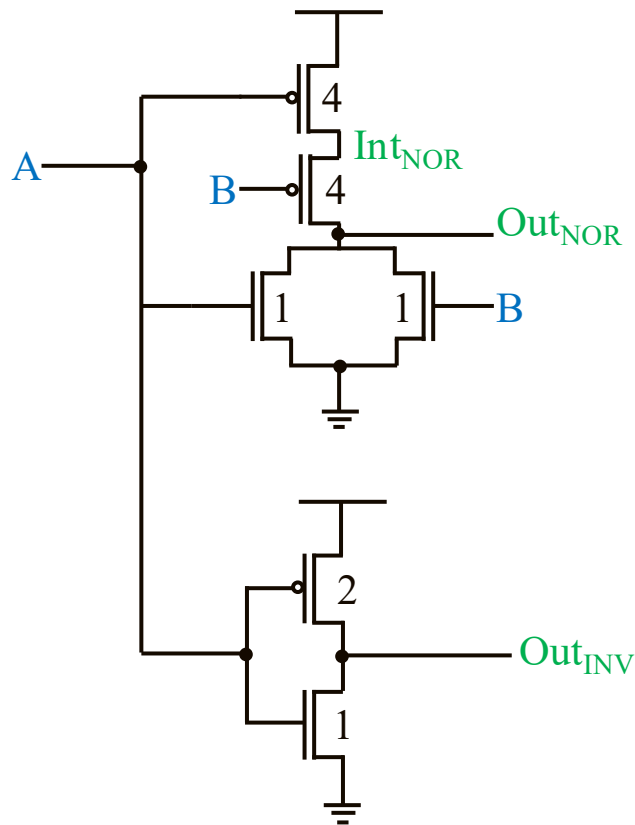
$$\tau_{pLH} = 8.4 \text{ pS}$$

Internode Delay



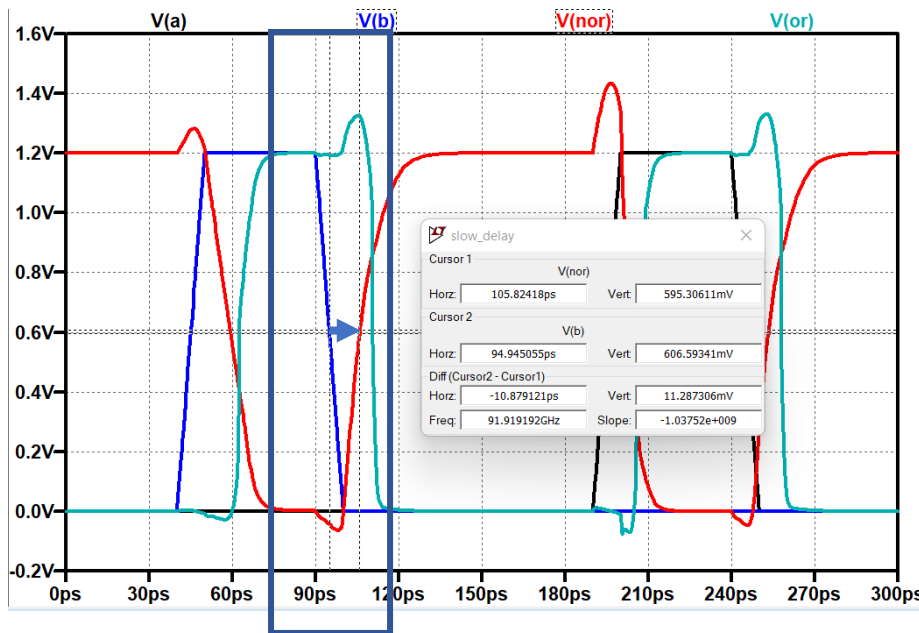
- Internode delay can be input dependent.
- Prior result of $\tau_{PLH} = 8.4$ ps was:
 - $B = 0$
 - $A = 1 \rightarrow 0$
- How does τ_{PLH} change for the following?
 - $B = 1 \rightarrow 0$
 - $A = 0$

Internode Delay

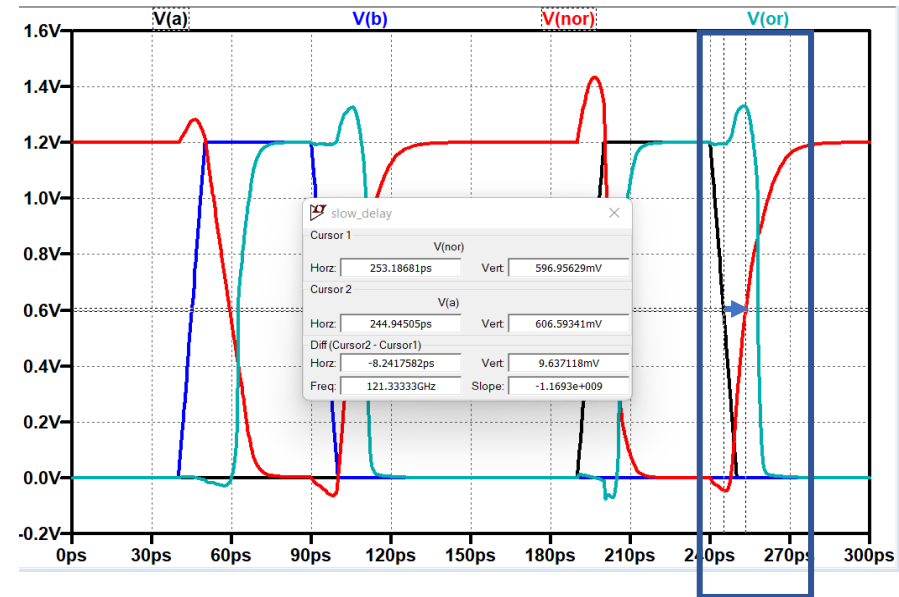


- Internode delay can be input dependent.
- Prior result of $\tau_{PLH} = 8.4$ ps was:
 - $B = 0$
 - $A = 1 \rightarrow 0$
- How does τ_{PLH} change for the following?
 - $B = 1 \rightarrow 0$
 - $A = 0$
- INT_{NOR} was already charged to V_{DD} .
 - Additional 2.1 ps of internode delay can be ignored.
 - Reduced delay of 6.3 ps.

Internode Delay – Input Dependence



- $\tau_{\text{PHL}(B=1 \rightarrow 0)} = 10.88 \text{ ps}$



- $\tau_{\text{PHL}(A=1 \rightarrow 0)} = 8.24 \text{ ps}$

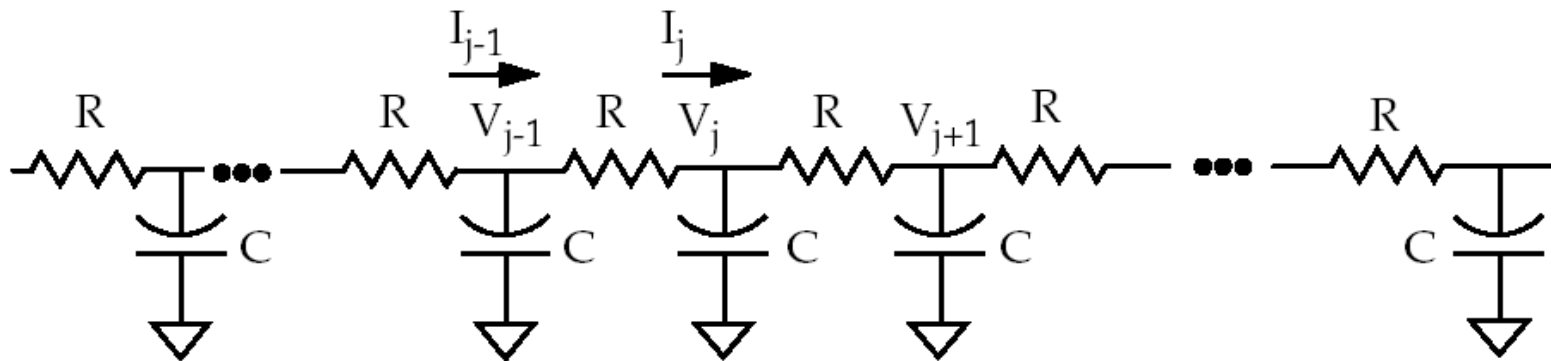
Internode Delay

- This method of inclusion of intermediate capacitances is known as Elmore delay.
- Elmore delay also encompasses capacitances that are not along the signal path, but must be charged as well.
- Let P_i denote the path from input to node i
- Let P_{ij} denote the portion of the paths P_i and P_j

$$\tau_{Di} = \sum_{j=1}^N C_j \sum_{k \in P_{ij}} R_k$$

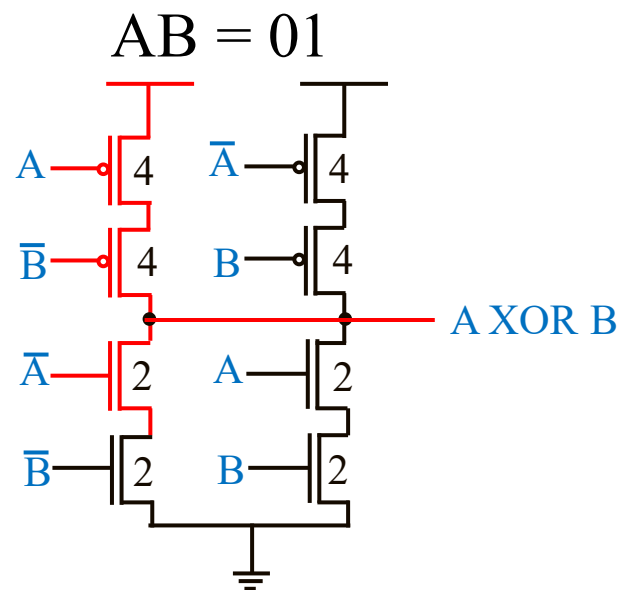
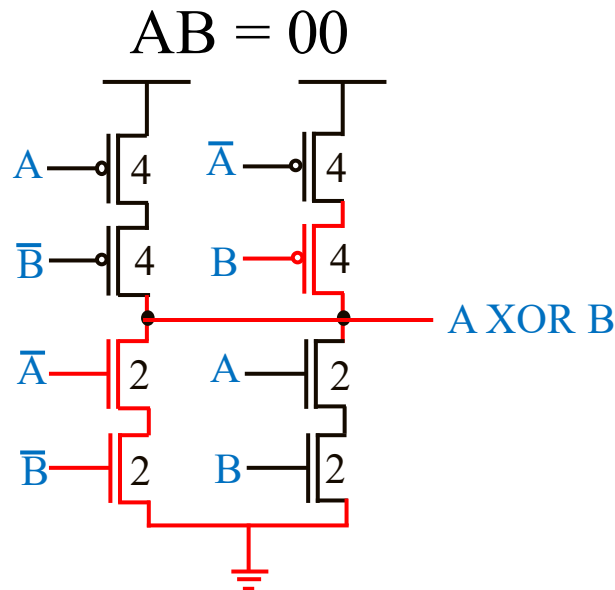
Internode Delay

- Elmore is a distributed RC model
- On-path delay
 - Each node along signal path must be charge
 - Charges with resistance R , which is the resistance along path to that point.



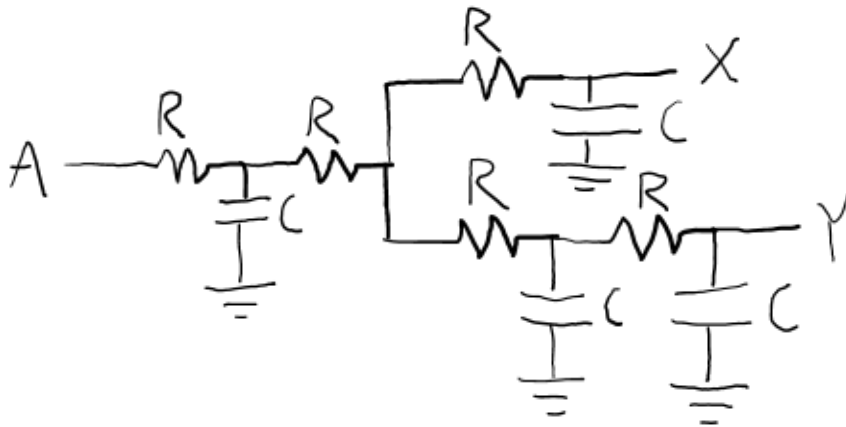
Elmore Delay – Off-paths

- RC branches off away from signal path still present a capacitance that is sinking current.
- EX: When $AB = 00 \rightarrow 01$,
 - $\bar{A}$ NMOS is on, presenting an off-path side capacitance that must be charged.



Elmore Delay – Off-paths

- We don't care how fast off-path capacitances charge.
 - Off-path R component is ignored.
 - Only R that is on-path until branch point.
 - Only C component is considered.



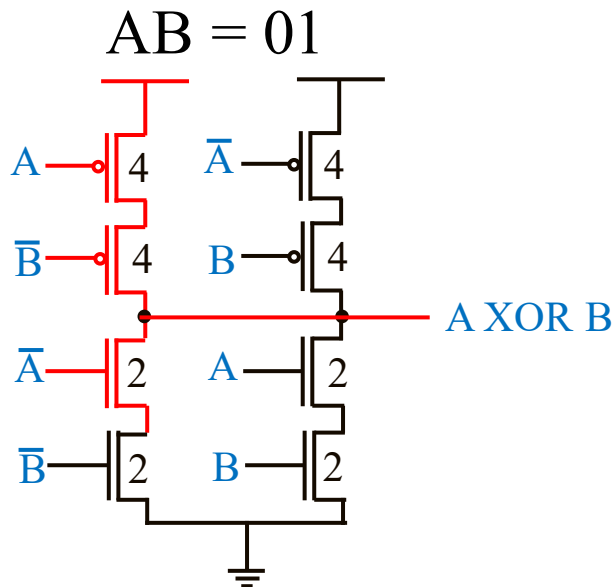
On-path delay $A \rightarrow Y$

$$RC + 3RC + 4RC = 8RC$$

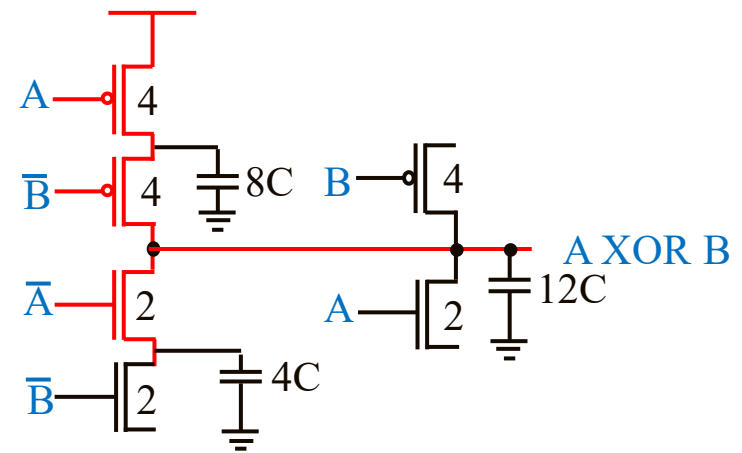
Off-path delay contribution:
 $2RC$

Elmore Delay – XOR

- EX: When $AB = 00 \rightarrow 01$,

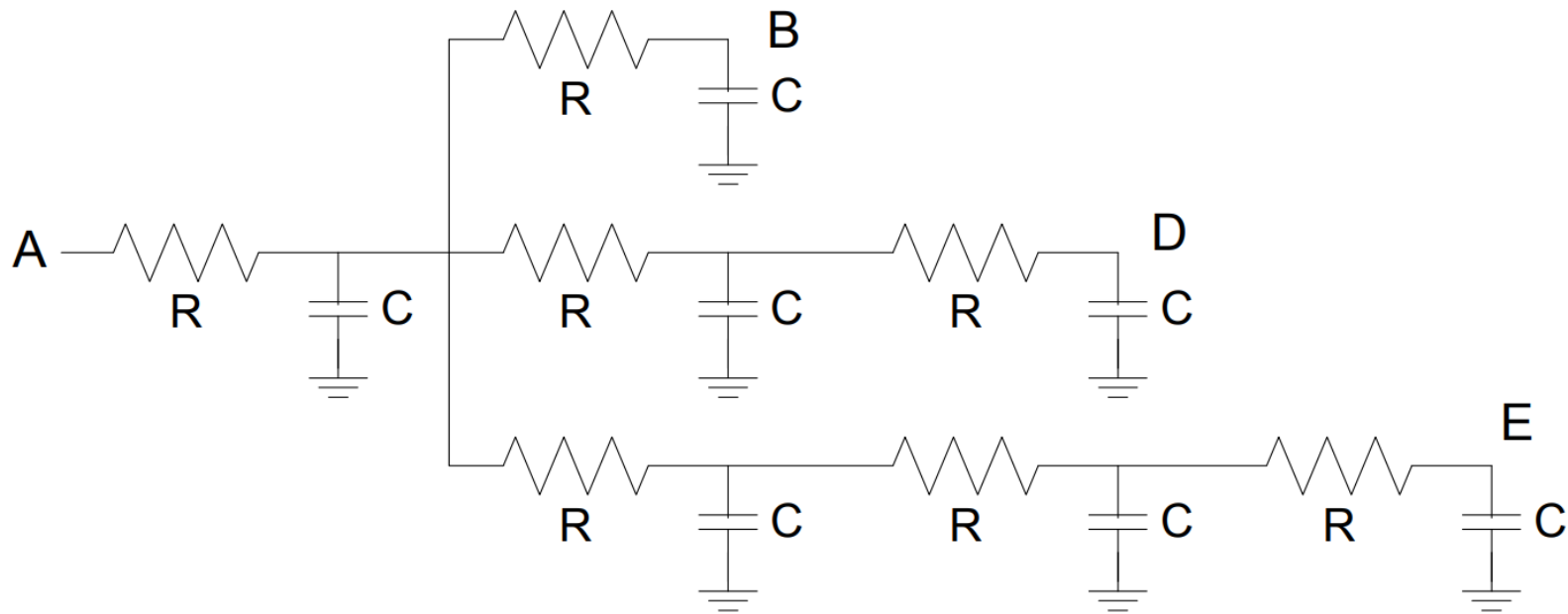


$$\tau = \underbrace{\frac{R_p}{4} \cdot 8C + \frac{R_p}{4} \cdot 2 \cdot 12C}_{\text{On-path PMOS delay}} + \underbrace{\frac{R_p}{4} \cdot 2 \cdot 4C}_{\text{Added delay from active NMOS}} = R_p C = 8 \cdot 30k \cdot 0.1f = 24ps$$



- All capacitances here are transistor output capacitances, with 0.1fF per 100nm of transistor width (C_{self} , or C_{output}).

Elmore Delay - Example



Elmore Delay - Example

τ_{AB} off-path on-path
 $5RC$ $RC + 2RC$, $\tau_{AB} = 5RC + 3RC$

 $\tau_{AB} = 8RC$

τ_{AD} off-path on-path
 $4RC$ $RC + 2RC + 3RC$, $\tau_{AD} = 4RC + 6RC$

 $\tau_{AD} = 10RC$

τ_{AE} off-path on-path
 $3RC$ $RC + 2RC + 3RC + 4RC$, $\tau_{AE} = 3RC + 10RC$

 $\tau_{AE} = 13RC$

Current Gain

- Driving a large C_L requires large Width.
- Large width increases C_L even more.
- Balance with cascade of inverters.

